

Newly Engineered Features for Phishing Detection

Explanations & Quantum-Inspired Connections

Phase 1 — Structural Features

Twelve structural features are derived from the raw URL string, domain, and JavaScript content. They capture syntactic irregularities that are hallmarks of phishing infrastructure.

1. `js_obfuscation_ratio` — Ratio of obfuscated JavaScript to total JavaScript

How it was added: Divides the length of obfuscated JavaScript (`web_js_obf_len`) by the total JavaScript length plus one to prevent division by zero:

$$\text{js_obfuscation_ratio} = \frac{\text{web_js_obf_len}}{\text{web_js_len} + 1}$$

Why it helps detect phishing

Phishing pages frequently hide malicious logic by obfuscating their JavaScript (e.g. encoding, minifying, or scrambling variable names). A high ratio of obfuscated to total JS is a strong indicator of deceptive web content.

Quantum connection

This ratio can be encoded as a rotation amplitude on a qubit. A high obfuscation ratio pushes the qubit toward one basis state, and a QML model can exploit this encoding alongside other features through entanglement to detect hidden malicious patterns.

2. `url_has_ip` — Whether the URL contains an IP address instead of a domain name

How it was added: Applies a regex pattern (`(\d{1,3}\.\d{1,3}\.\d{1,3}\.\d{1,3})`) to the URL string, returning 1 if an IP address is found, 0 otherwise.

Why it helps detect phishing

Legitimate websites almost always use human-readable domain names. Using a raw IP address in a URL is a classic phishing tactic to hide the true identity of a server.

Quantum connection

As a binary feature (0 or 1), this maps directly to a qubit basis state — $|0\rangle$ or $|1\rangle$. In a QML circuit it can act as a control qubit to conditionally gate operations on other URL-related features.

3. URL character-count features — `url_num_dots`, `url_num_hyphens`, `url_num_underscores`, `url_num_slashes`, `url_num_queries`, `url_num_ampersands`

How they were added: Each is a simple character count applied to the raw URL string, counting occurrences of `.`, `-`, `_`, `/`, `?`, and `&` respectively.

Why it helps detect phishing

Phishing URLs are often structurally abnormal — they may have excessive dots (deep subdomains), hyphens to imitate legitimate brands (e.g. `paypal-secure.com`), or long query strings to confuse users and bypass filters.

Quantum connection

These integer-valued features can be normalised and encoded as rotation angles $R_y(\theta)$ on individual qubits. A QML model can detect unusual combinations of these character counts through multi-qubit entanglement.

4. `url_suspicious_chars` — Count of suspicious characters in the URL

How it was added: Counts occurrences of known deceptive characters and character sequences: `@`, `~`, `%20`, `..`, `//`.

$$\text{url_suspicious_chars} = \#(\text{@}) + \#(\text{~}) + \#(\text{\%20}) + \#(\text{..}) + \#(\text{//})$$

Why it helps detect phishing

These characters are often used to disguise the true destination of a URL or inject path-traversal tricks. For example, `http://legit.com@evil.com` redirects to `evil.com`, not `legit.com`.

Quantum connection

Can be encoded as a qubit amplitude where higher counts push toward the phishing basis state. QML interference effects can amplify this signal when combined with other URL features.

5. `domain_num_subdomains` — Number of subdomains in the URL

How it was added: Extracts the domain from the URL and splits it by `.`; the subdomain count equals $\max(0, \text{number of parts} - 2)$.

Why it helps detect phishing

Phishing sites often stack multiple subdomains to appear legitimate (e.g. `secure.paypal.account.evil.com`). Legitimate domains rarely need more than one subdomain.

Quantum connection

This ordinal feature can be encoded as a multi-level rotation angle. QML models can capture non-linear thresholds (e.g. "more than 2 subdomains is very suspicious") that classical linear models may miss.

6. `domain_contains_numbers` — Whether the domain contains digits

How it was added: Extracts the domain name from the URL, then checks for any digit using the regex `\d`. Returns 1 if found, 0 otherwise.

Why it helps detect phishing

Legitimate brand domains almost never contain numbers (e.g. `paypal1.com` mimicking `paypal.com`). Digits in a domain often indicate typosquatting or auto-generated phishing domains.

Quantum connection

As a binary feature it directly maps to a qubit state. Combined with `url_has_ip` and `domain_suspicious_tld`, these binary qubits can be entangled to jointly detect numerically-obfuscated domains.

7. `domain_suspicious_tld` — Whether the domain uses a TLD known for phishing

How it was added: Checks if the top-level domain is in a predefined list of commonly abused TLDs: `.tk`, `.ml`, `.ga`, `.cf`, `.gq`, `.xyz`, `.top`, `.work`, `.click`, `.link`, `.download`.

Why it helps detect phishing

Free or very cheap TLDs are disproportionately used for phishing because attackers can register domains at no cost and discard them quickly.

Quantum connection

Binary feature (0/1) that maps to a qubit basis state. Can be entangled with `domain_trust_score` to create a joint representation of how untrustworthy a domain is.

Phase 2 — Entropy Features

Shannon entropy measures the information density (or randomness) of a string, capturing obfuscation and auto-generation patterns invisible to keyword searches.

$$H = - \sum_i p_i \log_2 p_i$$

8. `url_entropy` — Shannon entropy of the full URL string

How it was added: Computes character-level Shannon entropy across the entire URL string using the formula above.

Why it helps detect phishing

Long, random-looking URLs with high entropy (e.g. `http://x7f2k9.tk/a8Bq2Z`) are typical of auto-generated phishing links designed to evade pattern detection.

Quantum connection

Encodes directly as a qubit amplitude — high entropy pushes toward one basis state. A QML model can use this to distinguish random vs. structured URL patterns through quantum superposition.

9. `domain_entropy` — Shannon entropy of the domain portion only

How it was added: First extracts the domain via `urlparse`, then computes Shannon entropy on just the domain string.

Why it helps detect phishing

Complements `url_entropy` by isolating the randomness specifically within the domain name, where phishing obfuscation most commonly occurs (e.g. `a3kf9x.xyz`).

Quantum connection

While `url_entropy` captures global URL randomness, `domain_entropy` captures a finer signal. These two entropy qubits can be entangled to form a joint "URL structure" register in a QML circuit.

10. `content_entropy` — Shannon entropy of the first 1 000 characters of web content

How it was added: Truncates `web_content` to 1 000 characters for efficiency, then applies the Shannon entropy formula.

Why it helps detect phishing

Phishing pages that use encoded or template-generated content tend to have unusual entropy distributions compared to normal, human-written web content.

Quantum connection

Captures a signal orthogonal to URL-based entropy — the "information texture" of the page itself. Can be jointly encoded with `url_entropy` and `domain_entropy` as a three-qubit entropy register.

Phase 3 — Domain Trust Score

11. `domain_trust_score` — Composite trustworthiness score of the domain (0 to 1)

How it was added: Combines four weighted sub-signals clipped to $[0, 1]$:

$$\text{trust_score} = \underbrace{s_{\text{age}}}_{\in [0, 0.4]} + \underbrace{0.2 \cdot \mathbf{1}[\text{WHOIS complete}]}_{\in \{0, 0.2\}} + \underbrace{0.2 \cdot \mathbf{1}[\text{HTTPS}]}_{\in \{0, 0.2\}} + \underbrace{s_{\text{TLD}}}_{\in \{-0.1, 0, 0.2\}}$$

where the age score is:

$$s_{\text{age}} = \begin{cases} 0.4 & \text{if age} > 3 \times 365 \text{ days} \\ 0.3 & \text{if age} > 365 \text{ days} \\ 0.2 & \text{if age} > 180 \text{ days} \\ 0.1 & \text{if age} > 30 \text{ days} \\ 0 & \text{otherwise} \end{cases}$$

Why it helps detect phishing

A domain that is new, hides its WHOIS info, lacks HTTPS, and uses a suspicious TLD is almost certainly malicious. Combining these signals into one score captures multi-factor trust holistically.

Quantum connection

This multi-component score can be encoded as a controlled rotation angle. Since it captures feature interactions ($\text{age} \times \text{HTTPS} \times \text{TLD}$), a QML circuit can represent these joint dependencies through entangled qubits — something classical linear models struggle with.

Phase 4 — Email–URL Consistency Features

12. `email_domain_matches_url` — Whether the sender's email domain exactly matches the URL domain

How it was added: Extracts the domain from `email_from_domain` and the domain from `web_url`, then compares them exactly. Returns 1 if they match, 0 otherwise.

Why it helps detect phishing

A phishing email sent from `support@paypal-help.net` that links to `evil-redirect.com` is a clear domain mismatch. Legitimate emails typically link to the same domain that sent the email.

Quantum connection

Binary qubit feature. Can be used as a control qubit to gate other consistency features in a QML circuit, creating conditional logic that is naturally representable in quantum

computing.

13. `email_url_domain_similarity` — Soft similarity score between email domain and URL domain (0 to 1)

How it was added: A three-tier comparison:

$$\text{similarity} = \begin{cases} 1.0 & \text{exact match} \\ 0.7 & \text{one domain contains the other} \\ \frac{|A \cap B|}{|A \cup B|} & \text{Jaccard similarity on character sets} \end{cases}$$

Why it helps detect phishing

Complements the binary match feature by catching near-miss typosquatting cases (e.g. `paypal.com` vs. `paypall.com`). A low but non-zero similarity often signals an impersonation attempt.

Quantum connection

As a continuous value in $[0, 1]$, this maps naturally to a qubit rotation angle $R_y(\theta)$. QML can detect the non-linear boundary between "similar enough to be legitimate" and "similar enough to deceive" — a nuance that linear classifiers miss.

Phase 5 — Content Analysis Features

14–16. `content_num_forms`, `content_num_inputs`, `content_num_scripts` — HTML tag counts

How they were added: Each counts lowercase occurrences of `<form`, `<input`, or `<script` in the `web_content` string.

Why it helps detect phishing

Phishing pages are credential-harvesting pages — they need forms and inputs to steal data. An unusually high number of forms/inputs relative to page size, or a high script count, signals a data-collection trap. Legitimate informational pages rarely have multiple forms.

Quantum connection

These count features can be normalised and jointly encoded as rotation angles on a multi-qubit register. Their interactions (many scripts + many forms + many inputs together) can be captured through entanglement, enabling a QML model to detect the combined "credential harvesting" pattern.

17. `content_suspicious_keywords` — Count of high-risk phishing keywords in page content

How it was added: Counts how many keywords from a predefined list appear in the lowercase page content: *verify*, *urgent*, *suspended*, *account*, *confirm*, *password*, *update*, *click*, *login*, *security*.

Why it helps detect phishing

These words are the vocabulary of social engineering. Phishing pages use them to create urgency and push users into handing over credentials. A page containing many of these simultaneously is a strong phishing signal.

Quantum connection

This integer count can be normalised into a rotation angle on a qubit. Combined with `content_num_forms` and `email_url_domain_similarity`, a QML circuit can detect the full "phishing ecosystem" pattern — suspicious keywords + harvesting structure + domain mismatch — through multi-qubit entanglement.

Phase 6 — Semantic Coherence

18. `semantic_coherence_score` — How semantically consistent the email domain, URL, and page content are with each other (0 to 1)

How it was added: Takes the email domain, URL, and the first 500 characters of web content. Converts each to TF-IDF vectors (max 50 features, English stopwords removed), then computes pairwise cosine similarities and averages them:

$$\text{coherence} = \frac{1}{\binom{n}{2}} \sum_{i < j} \frac{\mathbf{v}_i \cdot \mathbf{v}_j}{\|\mathbf{v}_i\| \|\mathbf{v}_j\|}$$

where $\mathbf{v}_i$ is the TF-IDF vector of text source i and $n = 3$.

Why it helps detect phishing

Legitimate communications are semantically consistent — an Amazon email links to Amazon content. Phishing emails often mix unrelated contexts (e.g. a "bank security alert" email that links to a gambling site). Low coherence across these three sources is a strong phishing signal.

Quantum connection

This is the multi-source generalisation of the `q_semantic_similarity` feature. While that feature computes similarity between subject and body only, this score spans email + URL + content — a richer, three-way semantic signal. In QML, this can be encoded as a joint multi-qubit rotation, where each pairwise similarity contributes a partial rotation — analogous to a three-body quantum interaction.

Phase 7 — Brand Consistency

19. `brand_consistency_score` — Whether the brand referenced in the email matches the brand in the page content (0, 0.5, or 1)

How it was added: Extracts brand keywords from `email_from_domain` and from the first 500 characters of `web_content` using a predefined brand list: *paypal*, *amazon*, *google*, *microsoft*, *apple*, *facebook*, *netflix*, *bank*, *ebay*, *linkedin*, *instagram*, *twitter*. Then:

$$\text{brand_consistency_score} = \begin{cases} 1.0 & \text{if brand sets overlap (consistent)} \\ 0.0 & \text{if brands present but different (mismatch)} \\ 0.5 & \text{if no brand detected in either side (ambiguous)} \end{cases}$$

Why it helps detect phishing

A classic phishing tactic is brand impersonation — an email pretending to be from PayPal that leads to a page with no PayPal branding (or worse, another brand's content). A mismatch score of 0.0 is a direct red flag for impersonation.

Quantum connection

This ternary feature (0, 0.5, 1) maps to three distinguishable qubit states or rotation angles. Combined with `email_url_domain_similarity` and `semantic_coherence_score`, it forms a three-qubit "trust consistency" register in a QML circuit, allowing the model to detect the nuanced difference between impersonation, ambiguity, and legitimacy.